

CS & IT ENGINEERING



Computer Network

Transport Layer

Lecture No. - 11



By - Abhishek Sir



Recap of Previous Lecture



Topic

TCP Congestion Control



Topics to be Covered



Topic

TCP Window Scaling

Topic

TCP Timers

ABOUT ME



Hello, I'm **Abhishek**

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#Q. Suppose that the maximum transmit window size for a TCP connection is 12000 bytes. Each packet consists of 2000 bytes. At some point of time, the connection is in slow-start phase with a current transmit window of 4000 bytes. Subsequently, the transmitter receives two acknowledgments Assume that no packets are lost and there are no time-outs. What is the maximum possible value of the current transmit window?

[GATE-2004]

(A) 4000 bytes

✓ (B) 8000 bytes

(C) 10000 bytes

(D) 12000 bytes

$$MSS = 2000 \text{ byte}$$

$$CWND = 4000 \text{ bytes} = 2 MSS$$

$$CWND = 8000 \text{ byte} = 4 MSS$$

Ans: B



Topic : Bandwidth Delay Product

- bandwidth delay product = $\text{Bandwidth} * \text{RTT}$
[maximum amount of data that can be in transit on a network]
- Receiver window size field is 16-bits long
- Maximum possible available space in receiver's receiving buffer
can be advertised without window scale is $(2^{16} - 1)$ bytes ←
- if bandwidth delay product is greater than $(2^{16} - 1)$ bytes
than window scaling is required



Topic : Window Scaling



→ Both TCP client and server can advertise their scale factor (shift count)
[during TCP handshake via SYN packet]

$$\rightarrow \text{scale_factor} = [2^{\text{shift_count}}]$$

$$\rightarrow \text{Available space in receiver's receiving buffer (RWND)} \\ = [\text{Advertised Window Size} * \text{scale_factor}] \text{ bytes}$$



Topic : Window Scaling



→ Maximum value for shift count can be 14 *

$$0 < \text{shift count} \leq 14$$

→ Maximum possible available space in receiver's receiving buffer
can be advertised with window scale is $(2^{16} - 1) * 2^{14}$ bytes ←

#Q. Assume that the bandwidth for a TCP connection is 1048560 bits/sec. Let α be the value of RTT in milliseconds (rounded off to the nearest integer) after which the TCP window scale option is needed. Let β be the maximum possible window size with window scale option. Then the values of α and β are.

[GATE-2015, Set-2, 2-Mark]
[IIT-K, H.W.]

- (A) 63 milliseconds, 65535×2^{14}
- (B) 63 milliseconds, 65535×2^{16}
- (C) 500 milliseconds, 65535×2^{14}
- (D) 500 milliseconds, 65535×2^{16}



Topic : TCP Timer



TCP uses five timers :

1. Retransmission Timer
2. Persistence Timer
3. Keepalive Timer
4. Time-waited Timer → (2MSL)
5. Delayed Timer



Topic : Retransmission Timer

V. Imp.



→ Used to control a lost or discarded segment

→ TCP creates a "retransmission timer" when it sends a segment

↓
→ TCP uses dynamic retransmission time,
which is different for each connection and may be changed during connection



Topic : TCP Time-out Value

SampleRTT :

- Measured time from segment transmission until ACK receipt
- SampleRTT will vary

EstimatedRTT :

Initially : EstimatedRTT = SampleRTT

$$\text{EstimatedRTT} = (1 - \alpha) * \text{EstimatedRTT} + \alpha * \text{SampleRTT}$$

where $\alpha = 0.125$



Topic : TCP Time-out Value

DevRTT :

$$\text{DevRTT} = (1 - \beta) * \text{DevRTT} + \beta * |\text{SampleRTT} - \text{DevRTT}|$$

where $\beta = 0.25$

↓↓↓

$$\text{Time-out Interval} = \text{EstimatedRTT} + 4 * \text{DevRTT}$$

#Q. Consider the following statements about the timeout value used in TCP.

- i. The timeout value is set to the RTT (Round Trip Time) measured during TCP connection establishment for the entire duration of the connection.
- ii. Appropriate RTT estimation algorithm is used to set the timeout value of a TCP connection.
- iii. Timeout value is set to twice the propagation delay from the sender to the receiver.

Which of the following choices hold?

- (A) (i) is false, but (ii) and (iii) are true
- (B) (i) and (iii) are false, but (ii) is true
- (C) (i) and (ii) are false, but (iii) is true
- (D) (i), (ii) and (iii) are false

[GATE-2007, 1-Mark]

IIT-K, H.W

#Q. Suppose two hosts use a TCP connection to transfer a large file. Which of the following statements is/are False with respect to the TCP connection?

- I. If the sequence number of a segment is m , then the sequence number of the subsequent segment is always $m+1$.
- II. If the estimated round trip time at any given point of time is t sec, the value of the retransmission timeout is always set to greater than or equal to t sec.
- III. The size of the advertised window never changes during the course of the TCP connection.
- IV. The number of unacknowledged bytes at the sender is always less than or equal to the advertised window

[GATE-2015, Set-1, 1-Mark]

11T-K, H.W.

(A) III only

(B) I and III only

(C) I and IV only

(D) II and IV only



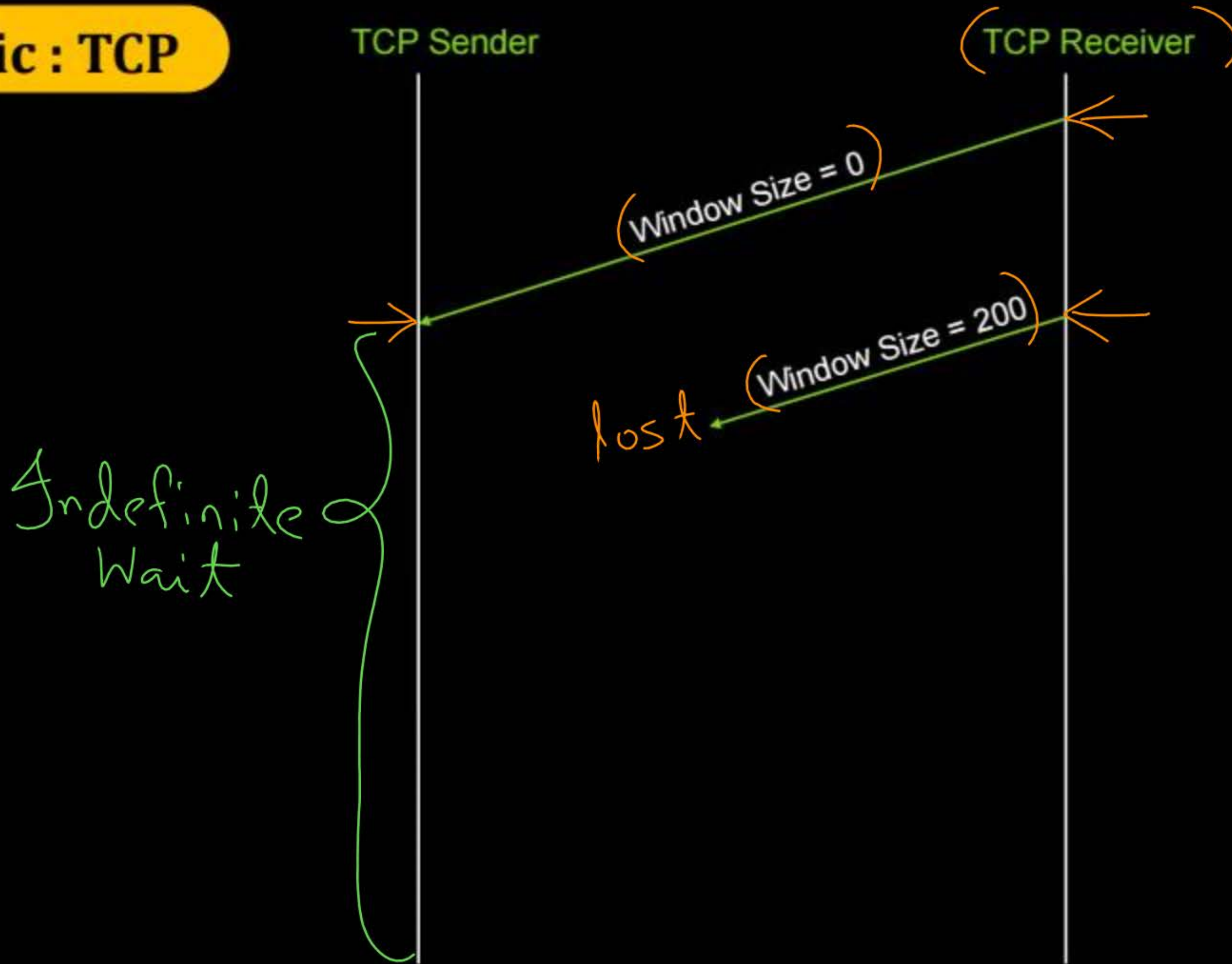
Topic : Persistence Timer



- When TCP receives a segment with a window size of "zero"
- TCP starts a persistence timer
- When the persistence timer goes off, the TCP sends a special segment called "probe"
- This prevents from deadlock



Topic : TCP





Topic : Keepalive Timer

- Prevents long idle connections between two TCPs
- Each time TCP hears from other TCP, it reset the timer
- if TCP does not hear from other TCP till keepalive timer time-out, the TCP sends a special segment called "probe"
- This prevents from indefinite wait ←



Topic : TCP



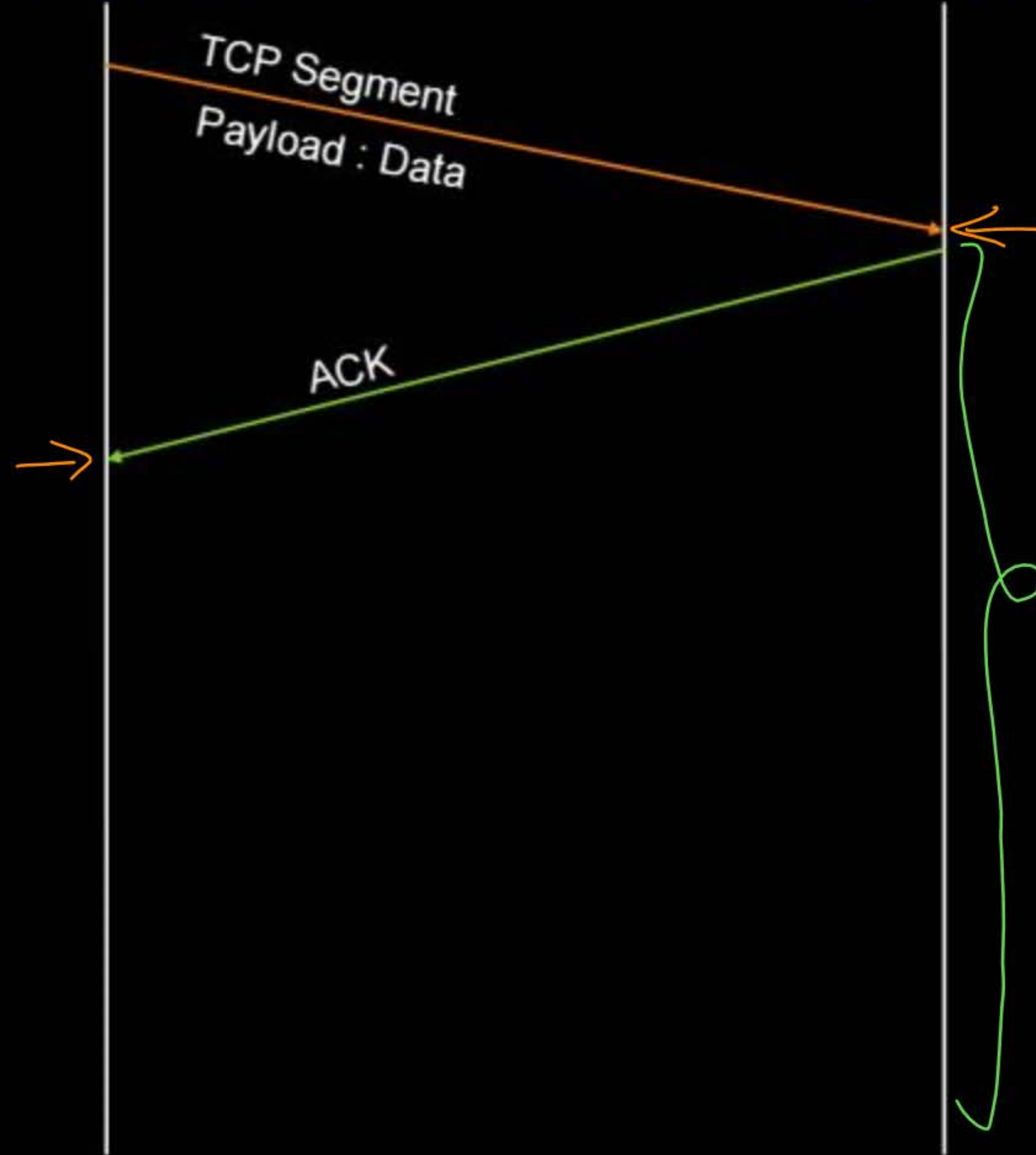
TCP Sender

TCP Receiver

TCP Segment
Payload : Data

ACK

Indefinite
Wait



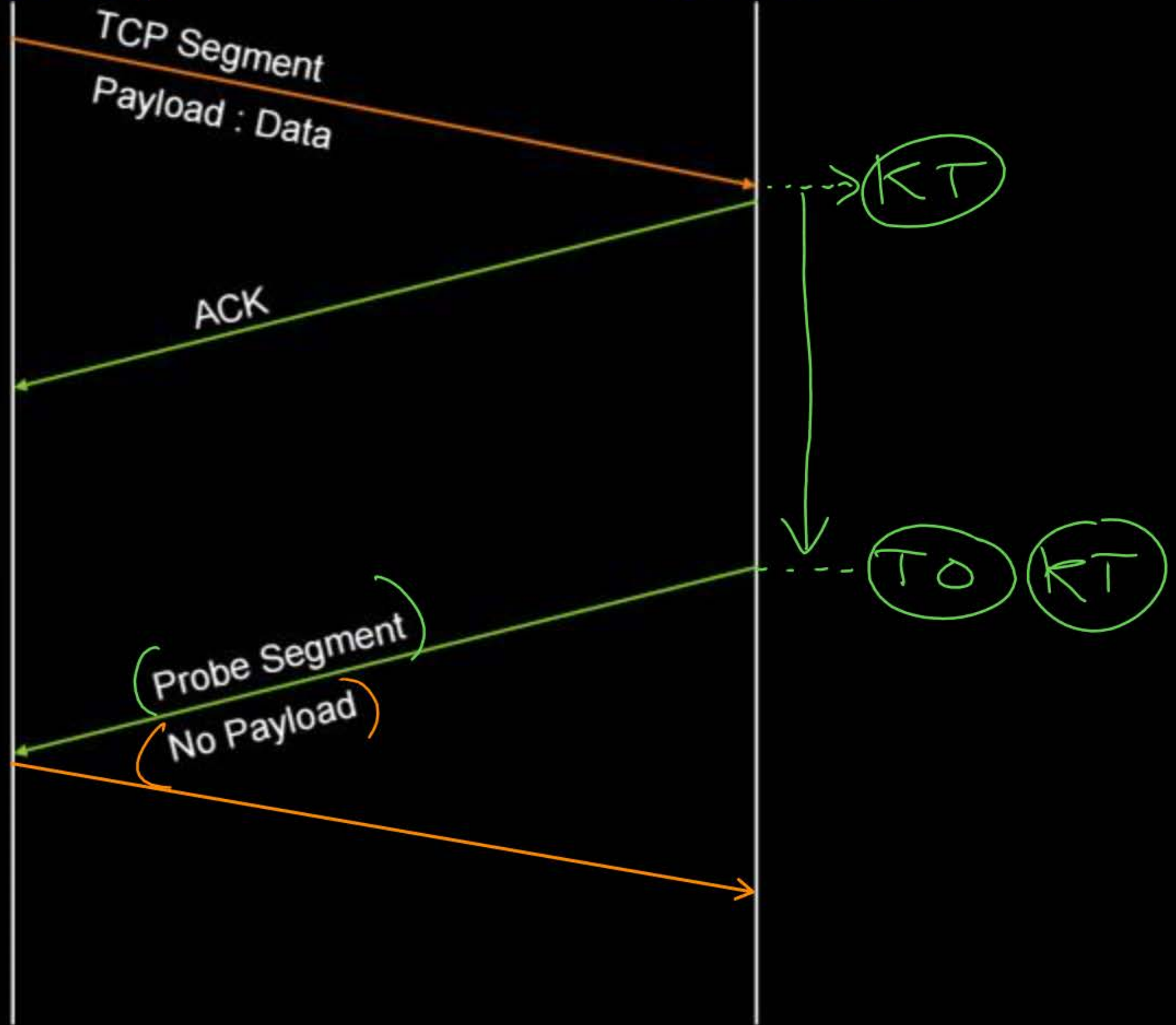


Topic : TCP



(TCP Sender)

TCP Receiver



[MSQ]

H.W.

IIT-B

[GATE-2021][2 Mark]



#Q. A TCP server application is programmed to listen on port number P on host S. A TCP client is connected to the TCP server over the network. Consider that while the TCP connection was active, the server machine S crashed and rebooted. Assume that the client does not use the TCP keepalive timer. Which of the following behaviors is/are possible?

- A If the client was waiting to receive a packet, it may wait indefinitely
- B The TCP server application on S can listen on P after reboot
- C If the client sends a packet after the server reboot, it will receive a RST segment
- D If the client sends a packet after the server reboot, it will receive a FIN segment



Topic : Time-waited Timer

- Used during connection termination (4-way handshake)
- More than twice the MSL (2MSL)
- When TCP closes a connection, it is not really closed, until this timer time-out
- Allow FIN segment to be received



Topic : Delayed Timer



- Method to optimize network traffic
[by delaying acknowledgments for received data segments]
- Instead of immediately sending an individual ACK for every segment
[allowing the ACK to piggyback on the next outgoing segment]
- TCP waits for a (short period) using delayed timer
[waiting for another segment will arrive, till timeout]

#Q. Which one of the following statements is FALSE?

[GATE-2004, 1-Mark]

IIT-D, H.W

- (A) TCP guarantees a minimum communication rate
- (B) TCP ensures in-order delivery
- (C) TCP reacts to congestion by reducing sender window size
- (D) TCP employs retransmission to compensate for packet loss

#Q. Packets of the same session may be routed through different paths in :

[GATE-2005, 1-Mark]

IIT-B, H.W.

- (A) TCP, but not UDP
- (B) TCP and UDP
- (C) UDP, but not TCP
- (D) Neither TCP, nor UDP



2 mins Summary



Topic

TCP Window Scaling ✓

Topic

TCP Timers ✓



THANK - YOU

